

SIM logic model and APDU

SIM (Subscriber Identity Module) is a UICC card used by mobile operators to provide users with services such as identity verification and network access. A SIM card stores a lot of data related to mobile network access, such as ICCID (UICC card ID), IMSI (International Mobile Subscriber Identity), authentication key (ki, Authentication key), LAI (Local Area Identity) and A list of emergency numbers related to the operator and a list of services that users can access. The SIM card also provides a PIN (Personal Identification Number) code for encrypting the SIM card and a PUK (Personal Unblocking Code) code for unlocking the PIN code to ensure the security of user data.

In the era of 2G networks, SIM cards include hardware and software. With the development of 3G mobile technology, SIM only contains software, but some of it is collectively called UICC (Universal Intergrated Circuit Card), and a new application USIM (Universal Subscriber Identity Module) has been added to the 3G network.

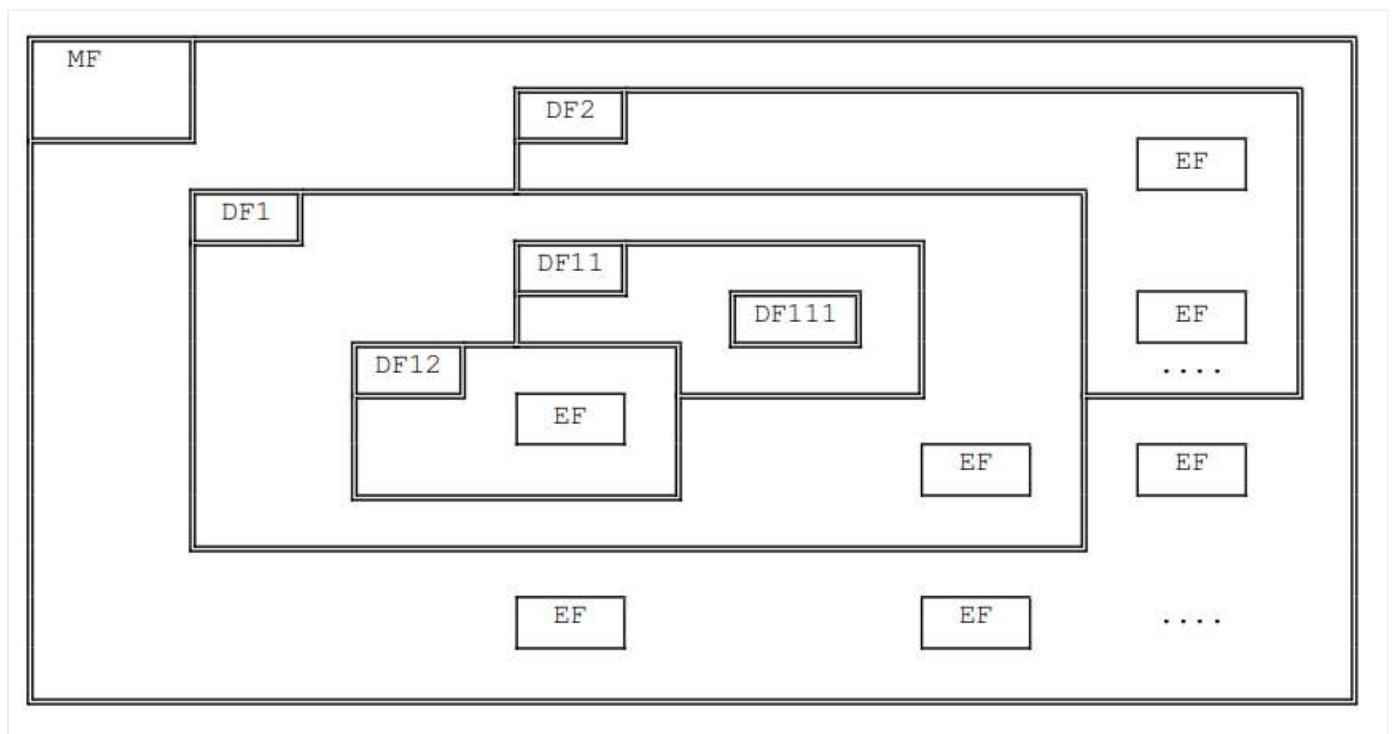
In this article, we will take a look at the logical structure of the SIM card, the stored data and the protocol APDU for reading the SIM card data.

UICC refers to a type of mobile terminal intelligent Smart cards are widely used in various mobile networks such as GSM, UMTS, and CDMA. UICC ensures the security and integrity of user data and provides users with services such as network access verification. In the GSM network, the SIM card is usually used; the UICC card in the UMTS network uses the USIM card. For services that support IMS (IP multimedia Services), the ISIM application needs to be included. For CDMA networks, the UICC card needs to include CSIM applications. The UICC card containing the three applications of SIM, USIM, and CSIM is called RUIM (Removable User Identity Card). The RUIM card can be inserted into mobile devices that support CDMA, GSM and UMTS. With the development of new technologies, a kind of Embedded SIM (embedded SIM, eSIM) cards began to be used by major operators, eSIM made the SIM card into a built-in chip embedded on the integrated circuit board to provide users with the corresponding network service.

The logical structure of SIM

The files in the SIM card are stored in a hierarchical structure (see the figure below), there are three main types: MF (Master File), DF (Dedicated File), EF (Elementary File) , The software system in the SIM card is

responsible for handling the access and writing of the file data.



The SIM file consists of two parts: header and body. The header is managed by the SIM card itself. It mainly contains information related to the file structure and attributes. It can be accessed using GET RESPONSE or STATUS (This information is fixed when the SIM card is generated); the body part contains the data contained in the corresponding file.

File Identifiers)

The file identifier (file id) is used to identify or address a specific file, and consists of two-byte hexadecimal numbers. For GSM SIM cards, the first byte is used to identify the file type:

- '3F': Master File (MF, main file);
- '7F': Dedicated File (DF, dedicated file);
- '2F': Elementary File under the Master File (EF, the basic file located under the MF);
- '6F': Elementary File under a Dedicated File (basic file located under DF).

The file identifier needs to meet the following conditions to ensure that each file can be accessed by a unique identifier:

- the file ID shall be assigned at the time of creation of the file concerned (the file ID shall be assigned when the relevant file is created);
- no two files under the same parent shall have the same ID (two files with a common father cannot have the same ID);

- a child and any parent, either immediate or remote in the hierarchy, eg grandparent, shall never have the same file ID (a child and any parent file, no matter in the hierarchy The direct or indirect relationship in the structure cannot have the same file ID).

Master File(MF)

There is only one main file MF in the SIM card, which is used to control the access of other proprietary files (DF) and basic files (EF).

Dedicated Files(DF)

The proprietary file is a functional file group, including itself and all sub-files. A DF file consists of only the header part:

- DFGSM which contains the application for both GSM and/or DCS1800;
- DFTELECOM which contains telecom service features.

The above two files are direct descendants of MF and can coexist in SIM cards that support multiple applications.

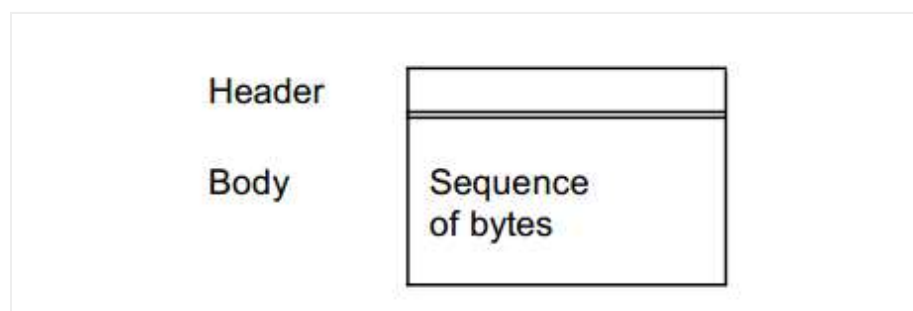
Elementary File(EF)

An Elementary File (EF) is composed of a header and a body part. The following three structures of an EF are used by GSM.

The basic file (EF) consists of two parts: header and body. In GSM, EF has the following three structure types:

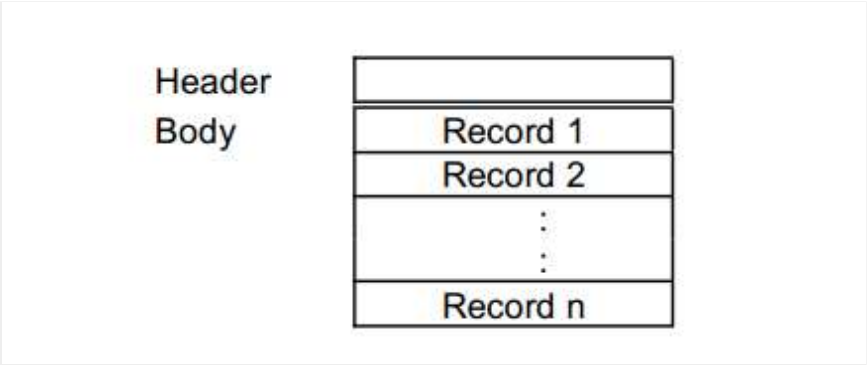
Transparent EF

The transparent structure of the EF file consists of a byte sequence. When reading or writing data, this byte sequence is accessed through a relative address (offset), which is used to indicate the starting position of the access and the number of bytes accessed. The relative address of the first byte of the transparent EF file is 00, And the EF file data length is included in the header part.



Linear Fixed EF

The linear fixed structure EF consists of a fixed-length record sequence, and the sequence number of the first record is 1. The length of a record and the length and the number of records are stored in the header part of EF.



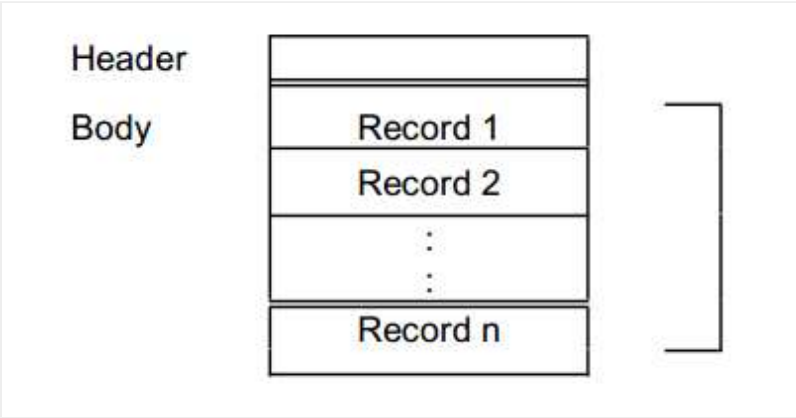
Note:

- It is not possible, at present, to have more than 255 records in a file of this type, and each record cannot be greater than 255 bytes (Currently EF files can only contain up to 255 Records; the length of each record cannot exceed 255 bytes).
- This structure was previously referred to as “formatted” in GSM (in GSM, this file was previously called a formatted file).

Cyclic EF

The loop file is used to store records arranged in time series. When all records are used for storage, new data will overwrite the oldest data.

The EF file with a loop structure is composed of fixed-length records, and each record can only contain fixed-length data. For this type of EF, there is a connection between the last record and the first record: when the record storage location points to the last record n, the next record is 1; in other words, when the next record pointer points to 1, its The last record is n. Record 1 contains the latest data, while record n contains the oldest data (the earliest recorded time).



Note: It is not possible, at present, to have more than 255 records in a file of this type, and each record cannot be greater than 255 bytes.

File selection method

After the ATR command, MF is selected by default and becomes the current directory. The selection of each file uses the SELECT function and follows the following principles:

ATR is a command message available for contact Smart Card. After the card is started up, it tells the reader that the card is ready and ready for use. reference:https://en.wikipedia.org/wiki/Answer_to_reset

- Select a DF or MF and set it as the current directory; after the selection, there is no current EF file;
- Select an EF, set the current EF, and the file directory is the parent DF or EF of the current EF. The current EF is always a subfile of the current directory.
- Any application-related instruction should be operable if the instruction is related to the current directory
- All selections are based on file ID

The following files can be obtained from the last selected file:

- Any direct subfile of the current directory;
- Any direct child DF file of the current DF father;
- the father of the current directory;
- current;
- MF.

The following table is the valid file selection path corresponding to the SIM logic structure model in the above figure:

Table 6: File selection

Last selected file	Valid Selections
MF	DF1, DF2, EF1
DF1	MF, DF2, DF3, EF2
DF2	MF, DF1, EF3, EF4
DF3	MF, DF1, EF5
EF1	MF, DF1, DF2
EF2	MF, DF1, DF2, DF3
EF3	MF, DF1, DF2, EF4
EF5	MF, DF1, DF3

Keep file ID

The following file IDs are reserved in GSM:

Proprietary file DF:

- administrative use (authorized use): '7F 4X';
- operational use:

'7F 10' (DFTELECOM), '7F 20' (DFGSM), '7F 21' (DFDCS1800); '7F 2X' ('2' <= x <= 'F').

Basic file EF:

- administrative use:

'6F XX' in the DFs '7F 4X';

'6F 1X' in the DFs '7F 10', '7F 20', '7F 21';

'2F 01', '2F EX' in the MF '3F 00';
- operational use:

'6F 2X', '6F 3X', '6F 4X' in '7F 10' and '7F 2X';

'2F 1X' in the MF '3F 00'.

▮ All of the above X, unless otherwise specified, their scope is ['0', 'F']

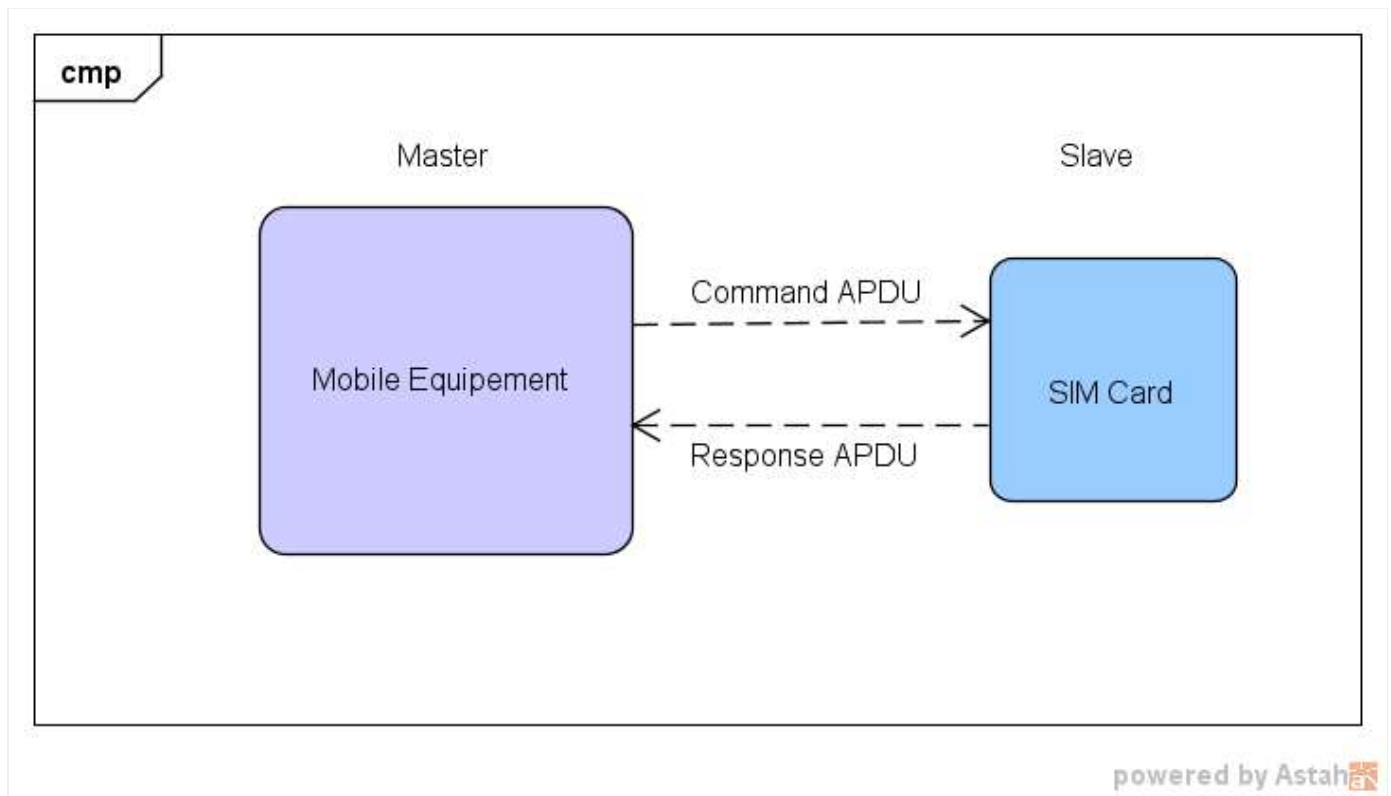
Basic file EF data

- EF-ICCID (2FE2): ICC Identification, the identity ID of the SIM card
- EF-LP (6F05): Language Preference, language preference
- EF-IMSI (6F07): International Mobile Subscriber Identity, the authentication ID for accessing the operator's network
- EF-KC (6F20): Ciphering Key kc, the encryption key for network registration
- EF-SST (6F38): SIM Service Table, a list of services provided by operators, such as ADN, FDN, SMS, etc.
- EF-ACM (6F39) : Accumulated Call Meter - total number of units for both the current call and the preceding calls
- EF-GID1 (6F3E) : Group Identifier Level 1 - identifiers for particular SIM-ME associations
- EF-GID2 (6F3F) : Group Identifier Level 2
- EF-SPN (6F46): Service Provider Name, operator name

- EF-MSISDN (6F40): Mobile Station International ISDN number
- (O)EF-CBML : Cell Broadcast Message Identifier Selection - Message format accepted by MS
- EF-BCCH (6F74) : Broadcast Control Channels
- EF-ACC (6F78) : Access Control Class - The access control class is a parameter to control the RACH utilisation.
- EF-FPMLN (6F7B) : Forbidden PMLNs
- EF-LOCI (6F7E): Location Information, location information
 - Temporary Mobile Subscriber Identity (TMSI)
 - Location Area Information (LAI)
 - TMSI TIME
 - Location update status
- EF-AD (6FAD) : Administrative Data - This EF contains information concerning the mode of operation according to the type of SIM, such as normal (to be used by PLMN subscribers for GSM operations), type approval (to allow specific use of the ME during type approval procedures of e.g. the radio equipment), cell testing (to allow testing of a cell before commercial use of this cell), manufacturer specific (to allow the ME manufacturer to perform specific proprietary auto-test in its ME during e.g. maintenance phases).
- EF-Phase (6FAE) : Phase Identification
- (O)EF-ADN (6F3A) : Abbreviative Dialing Number
- (O)EF-FDN (6F3B) : Fixed Dialing Number
- (O)SF-SMS (6F3C) : Short Message
- (O)EF-CCP (6F3D) : Capability Configuration Parameters - parameters of required network and bearer capabilities and ME configurations associated with a call established using an abbreviated dialling number, a fixed dialling number, MSISDN or a last number dialled.
- (O)EF-LND (6F44) : Last number dialled

APDU

The SIM card inserted in the mobile equipment (Mobile Equipment) has a master-slave relationship with the mobile phone itself. The ME exchanges data with the SIM card through the data exchange protocol APDU (Application Protocol Data Unit). The communication model between the two is shown in the figure below:



ME can access the SIM to read, update or delete the data in the SIM card through the following functional commands (only part of the commands are listed here):

- SELECT: select a file;
- STATUS: View the information of the current directory
- READ BINARY: Read byte sequence from transparent EF file
- UPDATE BINARY: Update the data of the current transparent EF file
- READ RECORD: Read the record of a linear fixed-length EF file or a circular EF file
- UPDATE RECORD: Update the record of linear fixed-length EF or circular EF file

How to map the above functions to the data exchange protocol APDU? In other words, the command format of the APDU must be determined. The communication mode between ME and SIM is half-duplex, that is, there are two communication channels, one is the request command channel, and the other is the data return channel, but the two cannot send data at the same time. Corresponding to this, APDU has two commands:

- Command APDU: The command that ME needs to send when requesting certain data from the SIM card;
- Response APDU: The APDU when the SIM responds to the ME's command request and returns to the ME.

Request APDU command consists of two parts: command header (required) and command body (optional), [TOP](#)

APDU Command						
Mandatory Head					Optional body	
CLA	INS	P1	P2	Lc	Data Field	Le

- CLA(1byte): Command category, used for application identification, used for GSM applicationsA0
- INS(1byte): request instruction code, such as SELECTA4; STATUS useF2
- P1(1byte)/P2(1byte): command parameter
- Lc: data length
- Le = 0(length of the expected response), if it is a write command, there is no data; if it is a read command, it needs to return 256 bytes of data
- Data Field: Data length is specified by Lc, byte sequence

The APDU of the request response consists of two parts: one part is data, and the other part contains the status code of the request command (return result of the request):

APDU Response		
Optional body	Mandatory part	
Data Field	SW1	SW2

- Data Field (can be changed into length): byte sequence
- SW1(1byte)/SW2(1byte): status code, indicating the result returned by the request command

The above SW1/SW2 status codes mainly include the following:

SW1	SW2	Meaning
0x90	0x00	Success
0x6E	0x00	CLA error
0x6D	0x00	INS error
0x6B	0x00	P1,P2 error
0x67	0x00	LEN error
0x98	0x04	Bad PIN
0x98	0x40	Card Blocked
0x98	0x02	no CHV initialized
0x94	0x00	no EF selected

For details about APDU, please refer to 3GPP TS 11.11 V8.14.0 Mobile Equipment(SIM-ME) interface

references

- <http://techiezjunkyard-sim.blogspot.kr/2011/12/logical-model-of-sim.html>
- 3GPP TS 11.11 V8.14.0 Mobile Equipment(SIM-ME) interface
- https://cedric.cnam.fr/~bouzefra/cours/Cartes_Intro_Anglais.pdf
- <http://www.tml.tkk.fi/Studies/T-110.497/2003/lecture4.pdf>

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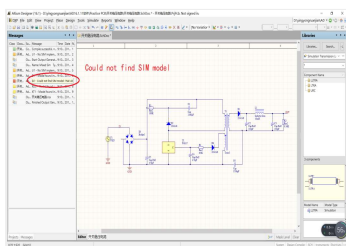
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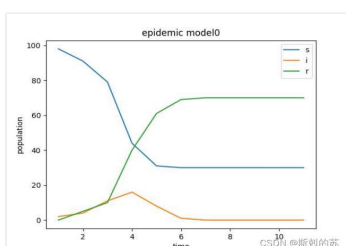
Golang tcp forwarding remoteAddr error

Intelligent Recommendation



The solution of could not find sim model appeared in AD simulation

In AD simulation, the error of could not find sim model sometimes appears. The solution is as follows: Double-click the error to locate the wrong component 2. Double-click the error component, select ...



DE-SIM example analysis (3) SIR infectious disease model

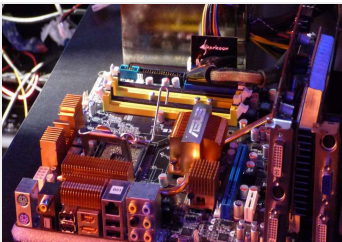
The first two examples of DE-SIM introduced one type of MESSAGE, multiple types of MESSAGE, a simulation object,

and the use of simulation objects. By setting the record frequency and record content t...



Add model to logic processing

In the business logic, you need to generate a model based on the data added by the user, instead of writing it in the py file under the model folder, that is, generating the model and model fields thr...



Model, simplified logic

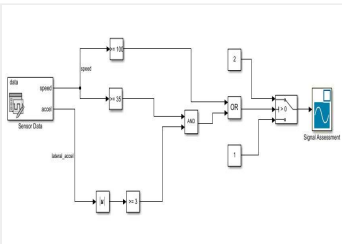
Learn FPGA, summarized as follows: Look logic built model. Only the establishment of a logical model in mind, understand the basis of FPGA internal logic structure implemented in order to understand w...

Business logic -Domain Model

Domain Model is the real world of business abstract class, so class includes not only data but also methods (real-world business logic). However, the art does not include a data access model, typicall...

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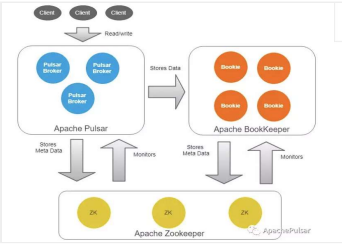
More Recommendation



Car performance logic model

Simulate the logic of a car performance package. There are dozens of controllers in the car to adjust the behavior of the

car in response to the data read by hundreds of sensors. In this simplified sy...



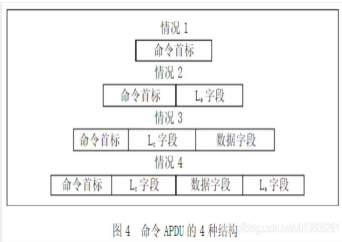
PULSAR logic storage model

1、BookKeeper Bookkeeper will be discussed in the background of Apache Pulsar Although Bookkeeper is a universal distributed log storage solution. First, Bookkeeper stores the data to the node in the c...

	Exam 1	Exam 2	Admitted
0	34.623660	78.024693	0
1	30.286711	43.894998	0
2	35.847409	72.902198	0
3	60.182599	86.308552	1
4	79.032736	75.344376	1

Logic regression model case

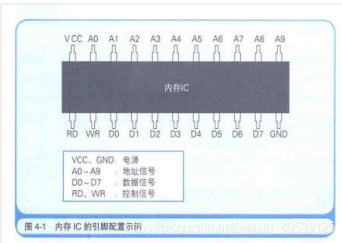
Logic regression model case We will build a logical regression model to predict whether a student is accepted by university. Suppose you are an administrator of a university department, and you want t...



Analysis of APDU instruction in ISO7816 protocol

Analysis of APDU instruction in ISO7816 protocol 1. APDU instruction structure 2. Decoding convention 3. APDU command format 4. APDU response 5.CLA 6.INS 7.P1,P2

8.SW2,SW2 1. APDU instruction structur...



Memory physical mechanism and logic model

Physical mechanism The above figure: VCC and GND are power supplies A0 ~ A9 is the pin of the address signal D0 ~ D7 is the pin of the data signal RD and WR are pins of the

control signal. After conne...

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